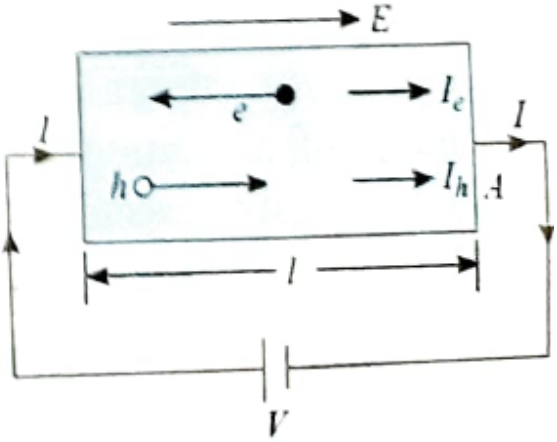
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In figure, a battery of emf $2\ \mathrm{V}$ is used. The length of the block is $0.1\ \mathrm{m}$ and the area is $1 \times 10^{4} \mathrm{m}^{2}$. If the block is of intrinsic silicon at $300\ \mathrm{K}$, find the electron and hole currents. What will be magnitude of the total current? What will be magnitude of the total current if germanium is used instead of silicon?
Given for silicon, $n = 1.5 \times 10^{16}/\mathrm{m}^{3}, \mu_e = 0.135\ \mathrm{m}^{2}/\mathrm{V_s}$,
$n = 2.4 \times 10^{19}/\mathrm{m}^{3}$
$\mu_h = 0.048\ \mathrm{m}^{2}/\mathrm{V_s}$ and for germanium,
$\mu_e = 0.39\ \mathrm{m}^{2}/\mathrm{V_s}$ and $\mu_h = 0.19\ \mathrm{m}^{2}/\mathrm{V_s}$



(A) $4.45 \times 10^{-3}\ \mathrm{A}$

(B) $4.45 \times 10^{-5}\ \mathrm{A}$

(C) $4.45 \times 10^{-2}\ \mathrm{A}$

(D) $5 \times 10^{3}\ \mathrm{A}$

Q.2

An experiment is performed to determine the $I - V$ characteristics of a Zener diode, which has a protective resistance of $\mathrm{R} = 100\Omega$, and a maximum power of dissipation rating of $1\ \mathrm{W}$. The minimum voltage range of the DC source in the circuit is:

(A) $0 - 5\ \mathrm{V}$

(B) $0 - 24\ \mathrm{V}$

(C) $0 - 12\ \mathrm{V}$

(D) $0 - 8\ \mathrm{V}$

Q.3

The grid voltage of any triode valve is changed from -1 volt to -3 volt and the mutual conductance is $3 \times 10^{-4}\mathrm{mho}$. The change in plate circuit current will be

(A) $0.8\ \mathrm{mA}$

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(B) $0.6\ \mathrm{mA}$

(C) $0.4\ \mathrm{mA}$

(D) $1\ \mathrm{mA}$

Q.4

Which one of the following statement is false ?

(A) Pure Si doped with trivalent impurities gives a p-type semiconductor

(B) Majority carriers in a n-type semiconductor are holes

(C) Minority carriers in a p-type semiconductor are electrons

(D) The resistance of intrinsic semiconductor decreases with increase of temperature

Q.5

In a triode, $g_m = 2 \times 10^{-3} \mathrm{ohm}^{-1}; \mu = 42;$ resistance
of load, $R = 50$ kilo ohm. The voltage amplification obtained from this triode will be

(A) 30.42

(B) 29.57

(C) 28.18

(D) 27.15

Q.6

The electric conductivity of an intrinsic semiconductor increases when the electromagnetic waves of wavelength equal or shorter than 2475 nm is incident on it. The charge carrier concentration of this semiconductor at room temperature is $1.6 \times 10^{16}\ \mathrm{m}^{-3}$ and the mobilities of electrons and holes at the same temperature are $0.4\ \mathrm{m}^2\ \mathrm{V}^{-1}\ \mathrm{S}^{-1}$ and $0.2\ \mathrm{m}^2\ \mathrm{V}^{-1}\ \mathrm{S}^{-1}$ respectively.
Regarding these information the band gap B is

(A) $B = 0.5\mathrm{eV}$

(B) $B = 1.3\mathrm{eV}$

(C) $B = 154 \times 10^{-3}\mathrm{eV}$

(D) None of these

Q.7

A transistor is connected in common -emitter configuration. The collector supply is $8\ \mathrm{V}$ and the voltage drop across a resistor of 800Ω in the collector is $0.5\ \mathrm{V}$. If the current gain factor α is 0.96. Find the base current.

(A) $26\mu\mathrm{A}$

(B) $16\mu\mathrm{A}$

(C) $10\mu\mathrm{A}$

(D) $5\mu\mathrm{A}$

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Q.8

Copper, a monovalent, has molar mass $63.54\ \mathrm{g/mol}$ and density $8.96\ \mathrm{g/cm^3}$. What is the number density n of conduction electron in copper?

(A) $3.2 \times 10^{20}\ \mathrm{m^{-3}}$

(B) $8.49 \times 10^{28}\ \mathrm{m^{-3}}$

(C) $6.2 \times 10^{31}\ \mathrm{m^{-3}}$

(D) None

Q.9

A pnp transistor is used in common-emitter mode in an amplifier circuit. A change of $40\mu\mathrm{A}$ in the base current brings a change of $2\ \mathrm{mA}$ in collector current and $0.04\ \mathrm{V}$ in base emitter voltage. Which of the following is the correct value of the input resistor R_{in} ?

(A) $\mathrm{R_{in}} = 9\mathrm{k}\Omega$

(B) $\mathrm{R_{in}} = 75\mathrm{k}\Omega$

(C) $\mathrm{R_{in}} = 1\mathrm{k}\Omega$

(D) None of these

Q.10

A potential barrier of 0.3 Volt exists across a p-n junction. The depletion region is $1\mu\mathrm{m}$ wide. The intensity of electric field in this region is E. If electron with speed $5 \times 10^5\ \mathrm{m/s}$ approaches this p -n junction from n -side, what will be its speed V on entering the p-side?

(A) $\mathrm{V} = 12 \times 10^5\ \mathrm{m/s}$

(B) $\mathrm{V} = 1 \times 10^4\ \mathrm{m/s}$

(C) $\mathrm{V} = 3.8 \times 10^5\ \mathrm{m/s}$

(D) $\mathrm{V} = 1.6 \times 10^4\ \mathrm{m/s}$

Q.11

The ratio of work function and temperature of two emitters are $1:2$, then the ratio of current densities obtained by them will be

(A) $4:1$

(B) $2:1$

(C) $1:2$

(D) $1:4$

Q.12

The bit rate for a signal, which has a sampling rate of $8\mathrm{kH}z$ and where 16 quantisation levels have been used is

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(A) 32000 bits/s

(B) 16000bits/s

(C) 64000 bits/s

(D) 72000 bits $/\text{s}$

Q.13

In frequency modulation

(A) The amplitude of modulated wave varies as frequency of caxrier wave

(B) The frequency of modulated wave varies as amplitude of modulating wave

(C) The frequency of modulated wave varies as frequency of modulating wave

(D) The frequency of modulated wave varies as frequency of carrier wave

Q.14

A piece of copper and the other of germanium are cooled trom the room temperature to $80\ \text{K}$, then which of the following would be a correct statement?

(A) Resistance of each increases

(B) Resistance of each decreases

(C) Resistance of copper increases while that of germanium decreases

(D) Resistance of copper decreases while that of germanium increases

Q.15

A junction diode of negligible forward resistance is used as a half-wave rectifier for a sinusoidal voltage supply $24\sqrt{2}$ V (RMS), 50 Hz. The reading of D.C. voltmeter connected across the capacitor of 22 μF which is in turn connected in parallel to the output load resistance R_L of 20 kΩ is _____ V.

(A) 48.25

(B) 48.50

(C) 48.75

(D) 48.00

Answers & Solutions

Q.1 Answer:

$4.45 \times 10^{-3}\ \text{A}$

Solution:

We are given that e.m.f of the battery $V = 2\ \text{V}, l = 0.1\ \text{m}, A = 1 \times 10^4\ \text{m}^2$
In case of silicon, $n_c = n_h = n = 1.5 \times 10^{16}/\text{m}^3$
Electric field, $E = \frac{V}{l}$ **where,**
e.m.f of the battery $V = 2\ \text{V}$,
and length of the block $l = 0.1\ \text{m}$

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$\Rightarrow \quad E = \frac{2\text{ V}}{0.1\text{ m}} = 20\text{ V/m}$

Clearly, $v_c = E\mu_c = 20 \times 0.135\text{ m/s} = 2.70\text{ m/s}$

$v_h = E\mu_h = 20 \times 0.048\text{ m/s} = 0.96\text{ m/s}$

Electron current, $I_c = e.4n_c v_c = \left(1.6 \times 10^{-19}\right)\left(10^{-4}\right)\left(1.5 \times 10^{16}\right) 2.70$

$= 6.48 \times 10^{-7}\text{ A}$

Hole current, $I_h = cAn_h v_h = \left(1.6 \times 10^{-19}\right)\left(10^{-4}\right)$

$= 2.304 \times 10^{-7}\text{ A}$

Total current, $I = I_e + I_h = \left(6.48 \times 10^{-7}\text{ A}\right) + ($

$= 8.784 \times 10^{-7}\text{ A}$

In case of germanium, $n_e = n_h = n = 2.4 \times 10^{19}/\text{m}^3$

$v_e = E\mu_e = 20 \times 0.39 = 7.8\text{ m/s}$

$v_h = E\mu_h = 20 \times 0.19 = 3.8\text{ m/s}$

Total current, $I = eA\left(n_e v_e + n_h v_h\right)$

$I = \left(1.6 \times 10^{-19}\right)\left(10^{-4}\right) \times \left[\left(2.4 \times 10^{19}\right)(7.8) + \left(2.4 \times 10^{19}\right)(3.8)\right]$

$\Rightarrow \quad I = 4.45 \times 10^{-3}\text{ A}$

Q.2 Answer:

$0 - 12\text{ V}$

Solution:

The minimum voltage range of DC **source is given by** $\text{V}^2 = \text{PR} \quad \because \text{P} = 1$ **watt,** $\text{R} = 100\Omega = 1 \times 100$

$\therefore \text{V} = 10$ **volt**

Q.3 Answer:

0.6 mA

Solution:

$$\text{g}_\text{m} = \frac{\Delta \text{I}_\text{p}}{\Delta \text{V}_\text{g}}$$

$$\Delta \text{I}_\text{p} = \text{g}_\text{m} \times \Delta \text{V}_\text{g} = 3 \times 10^{-4} \times [-3 - (-1)]$$

or $= -0.6 \times 10^{-3}\text{ A} =$ shortage of $0.6 \times 10^{-3}\text{ A}$

Q.4 Answer:

Majority carriers in a n-type semiconductor are holes

Solution:

Majority carriers in an n-type semiconductor are electrons.

Q.5 Answer:

29.57

Solution:

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$A_v = \frac{\mu R}{r_p + R} = \frac{\mu R}{(\mu / g_m) + R}$
$= \frac{42 \times \left(50 \times 10^3\right)}{42/(2 \times 10^{-3}) + 50 \times 10^3} = 29.57$

Q.6 Answer:

$\mathrm{B} = 0.5\mathrm{eV}$

Solution:

Forbidden energy gap $= \frac{12400(\mathrm{eV} - \mathring{A})}{\lambda(\mathring{A})}$
$= \frac{12400}{24750}\mathrm{eV} = 0.5\mathrm{eV}$

Q.7 Answer:

$26\mu\mathrm{A}$

Solution:

$\beta = \frac{a}{1-\alpha} = 24$
$i_C = \frac{0.5\mathrm{v}}{800\Omega} = 0.625 \times 10^{-3}\ \mathrm{A}$
$i_B = \frac{i_c}{\beta} = \frac{0.625 \times 10^{-3}}{24} = 26\mu\mathrm{A}$

Q.8 Answer:

$8.49 \times 10^{28}\ \mathrm{m}^{-3}$

Solution:

If M is the molar mass and ρ is the density then volume of one mole

$$V = \frac{M}{\rho}$$

The number of atoms per unit volume $= \frac{N_A}{V} = \frac{N_A}{M/\rho} = \frac{N_A \rho}{M} = \frac{\left(6.02 \times 10^{23}\right) \times (8.96)}{63.54}$
$= 8.49 \times 10^{22}\ \mathrm{cm}^{-3} = 8.49 \times 10^{28}\ \mathrm{m}^{-3}$

As each copper (monovalent) atom has one electron, so number of electrons per unit volume $= 8.49 \times 10^{26}\ \mathrm{m}^{-3}$

Q.9 Answer:

$\mathrm{R_{in}} = 1\mathrm{k}\Omega$

Solution:

Input resistance $R_{in} = \frac{\Delta V_{BE}}{\Delta i_B} = \frac{0.04}{40 \times 10^{-6}} = 1k\Omega$

Q.10 Answer:

$\mathrm{V} = 3.8 \times 10^5\ \mathrm{m/s}$

Solution:

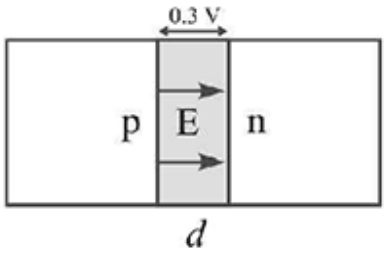
The intensity of electric field $E = \frac{V}{d} = \frac{0.3}{1 \times 10^{-6}} = 3 \times 10^5\ \mathrm{V/m}$

The electric field retarded the electron. The retardation

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$a = \frac{Ee}{m}$

The speed of the electron on entering p-side



$v^2 = u^2 - 2as = u^2 - 2\frac{eE}{m} \cdot d$
$= \left(5 \times 10^5\right)^2 - \frac{2 \times 1.6 \times 10^{-19} \times 3 \times 10^5}{9.1 \times 10^{-31}} \times \left(1 \times 10^{-6}\right)$
$= 3.8 \times 10^5 \text{ m/s}$

Q.11 Answer:

$1 : 4$

Solution:

$$\frac{\mathrm{J_1}}{\mathrm{J_2}} = \frac{\mathrm{AT_1^2 e^{-W_1/kT_1}}}{\mathrm{AT_2^2 e^{-W_2/kT_2}}}$$
$$= \left(\frac{\mathrm{T_1}}{\mathrm{T_2}}\right)^2 \mathrm{e}^{-\frac{\mathrm{W_1}}{\mathrm{kT_1}} + \frac{\mathrm{W_2}}{\mathrm{kT_2}}}$$
$= \frac{w_1}{kT_1} = \frac{w_2}{kT_2} \Rightarrow \left(\frac{T_1}{T_2}\right)^2 e^0$
$= \frac{1}{4}$

Q.12 Answer:

32000 bits/s

Solution:

If n is the number of bits per sample, then number of quantisation levels $= 2^{\mathrm{n}}$

Since the number of quantisation levels is 16

$\Rightarrow 2^n = 16 \Rightarrow n = 4$

$\therefore$ **bit rate = sampling rate** $\times$ **no. of bits per sample**

$= 8000 \times 4 = 32,000$ **bits** $/\mathrm{s}$

Q.13 Answer:

The frequency of modulated wave varies as amplitude of modulating wave

Solution:

In frequency modulation the frequency of the modulated wave is the linear function of the amplitude of the modulating wave.

Q.14 Answer:

Resistance of copper decreases while that of germanium increases

Solution:

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Resistance of conductors (Cu) decrease with decrease in temperature while that of semi-conductors (Ge) increase with decrease in temperature.

Q.15 Answer:

48.00

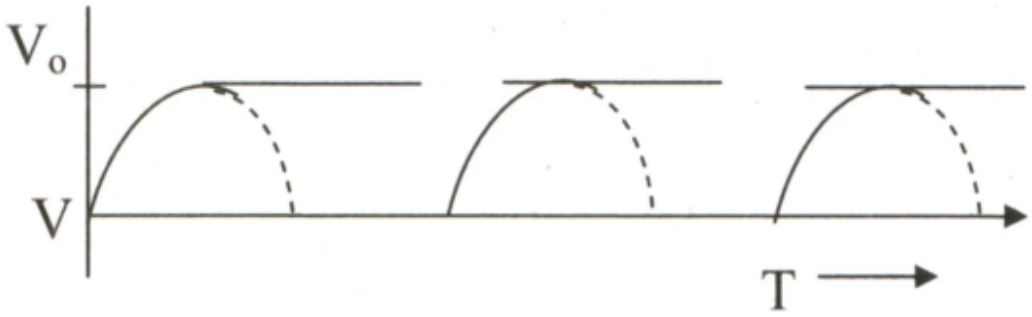
Solution:

When a capacitance is connected across R_L capacitor charges at peak supply and when diode is not conducting, it discharges through R_L. Now, time period of supply voltage,

$T = \frac{1}{f} = \frac{1}{50} = 0.02s$

Time constant of CR circuit, $CR = 22 \times 10^{-6} \times 20 \times 10^{3} = 0.44s$

since, $CR \gg T$, **the output voltage across capacitor will practically remain equal to the peak value of supply voltage and hence output will be as shown in figure.**



Hence, reading of d.c. voltmeter connected acrossC **is** V_o

Also, $V_{rms} = \frac{V_o}{\sqrt{2}}$

$\therefore \quad V_0 = V_{rms}\sqrt{2} = 24\sqrt{2} \times \sqrt{2} = 48V$

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